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Q3. Chloroplasts are the organelles where photosynthesis happens.

The main pigment is chlorophyll a and b, also carotenoids.

Stage 1 - Light reaction: captures light energy in the thylakoid.

Stage 2 - Dark reaction (Calvin cycle): uses that energy to make glucose in the stroma.

Q1. The artery carries blood away from the heart. The main one is the aorta.

Q2. The chloroplast is the organelle involved in photosynthesis.

Q5. [Diagram: alveolus drawn as a round sac with capillaries wrapped around it]
Labels: air space, alveolar wall, capillary, O_2 in, CO_2 out.
Gas exchange happens by diffusion across the thin wall.

Q8. Palisade mesophyll is at the top, cells are long and packed tightly with many chloroplasts, so it absorbs the most light.
Spongy mesophyll is below, cells are round with big air spaces between them, which lets gases diffuse easily to the stomata.

Q6. [Diagram: human digestive system]
Labelled - mouth, oesophagus, stomach, liver,
pancreas,
small intestine, large intestine, rectum.
The stomach churns food with acid and pepsin.

Q9. Transpiration is the loss of water vapour from the
leaves through the stomata.
Water moves up the xylem to replace it. Two factors
that increase the rate are
high temperature and moving air (wind).

...continued from Q6

Most absorption happens in the small intestine, in the ileum,

because of the villi which give a huge surface area.

11. Plant B was kept in dim light so it could not photosynthesise properly, that is why the leaves are pale and stretched out looking for light.
To help it recover, move Plant B to a bright spot with indirect sunlight and water it normally.

Q14. Rough work: $0.5 \times 12 = 6 \text{ L/min}$
 $0.5 - 0.15 = 0.35$, $0.35 \times 12 = 4.2 \text{ L/min}$